

The Development Application Base of Engineering Talents (iDABET)

Project of the Industrial Development Administration, Ministry of Economic Affairs

2025 Practical Ability Development Implementation Unit Project – Engineering Talent Practical Ability Development Record Form

Project No. of the Practical Ability Development Implementation Unit: DABET11403

No. of Engineering Talent: 40

Name of the Practical Ability Development Implementation Unit: 財團法人金屬工業研究發展中心/光電技術組

Project Name of the Practical Ability Development Implementation Unit: 半導體產業之真空設備與製程能力發展與人工智慧控制應用

Name of the Engineering Talent: 蘇翼

Item	Date	Instructor	No. of Hours (Note 1)	Type of Practical Ability Development Topic (Note 2)	Topic	Description
1	4/2	李宗信	4	1	Indoor Positioning System Research(Thesis)	Self Study: Developing a simulation for testing robot's movement
2	4/4	李宗信	5	1	Indoor Positioning System Research(Thesis)	Self Study: Developing a simulation for testing robot's movement
3	4/5	李宗信	3	1	Indoor Positioning System Research(Thesis)	Self Study
4	4/7	李宗信	8	1	Indoor Positioning System Research(Thesis)	Image Processing Workshop (Self Study)
5	4/9	李宗信	8		Indoor Positioning System Research(Thesis)	Object Recognition Workshop (Self Study)
6	4/10	顏仲崑老師	2	2	CNC Machining	Learning the fundamentals of the CNC milling machine
7	4/10	顏仲崑老師	3	1	CNC Machining	Learning the fundamentals of the CNC milling machine
8	4/11	李宗信	8	1	Indoor Positioning System	Machine Learning Workshop (Self Study)

					Research(Thesis)	
9	4/17	李宗信	8	1	Indoor Positioning System Research(Thesis)	Self-Study
10	4/18	李宗信	4	1	Indoor Positioning System Research(Thesis)	Self-Study
11	4/21	李宗信	8	1	Indoor Positioning System Research(Thesis)	Self-Study
12	4/22	李宗信	6	1	Indoor Positioning System Research(Thesis)	Self-Study
13	4/24	李宗信	2	2	PVD Coating Technology	The Pros and Cons of Physical and Chemical Vapor Deposition
14	4/24	洪忠欽/孫榮祐	2	1	X-ray Technology and Applications of PVD in the Semiconductor Industry	Applications of X-ray for checking defects in products.
Total Hours on Practical Ability Development Topics			Total ____71____ hours.			

Please add rows to the table as needed.

- Notes: (1) The hours for implementing practical ability development topics should be filled in according to the actual implementation status between March 1 and August 31.
- (2) Type of practical ability development topics: 1. Practice-based; 2. Industry collaboration-based.
- (3) Please attach at least 6 dated photos (yyyy/mm/dd) as evidence of the engineering talent's participation in practical ability development topics or activities.
- (4) Both the instructor and engineering talent shall sign at the bottom of the record form.
- (5) To implement the “2025 Development Application Base of Engineering Talents (iDABET) Project”, the Industrial Development Administration, Ministry of Economic Affairs will collect, process and use the personal data of the party involved without notifying the said party, as prescribed in Subparagraph 2 of Paragraph 2 of Article 8 of the Personal Data Protection Act.

【Photos of Participation in Practical Ability Development Topics or Activities】



On this day, I attended a Deep Learning workshop that introduced the basics of Python, object detection, image classification, and machine learning. I am now able to grasp these concepts and incorporate them into my project.



In this class, I learned the basic commands of a CNC machine through Siemens virtual simulation. We calibrated the tool relative to the workpiece to ensure precise cutting. This experience gave me valuable insight into how I can apply similar principles when programming my hexapod robot to set accurate leg positions during motion.



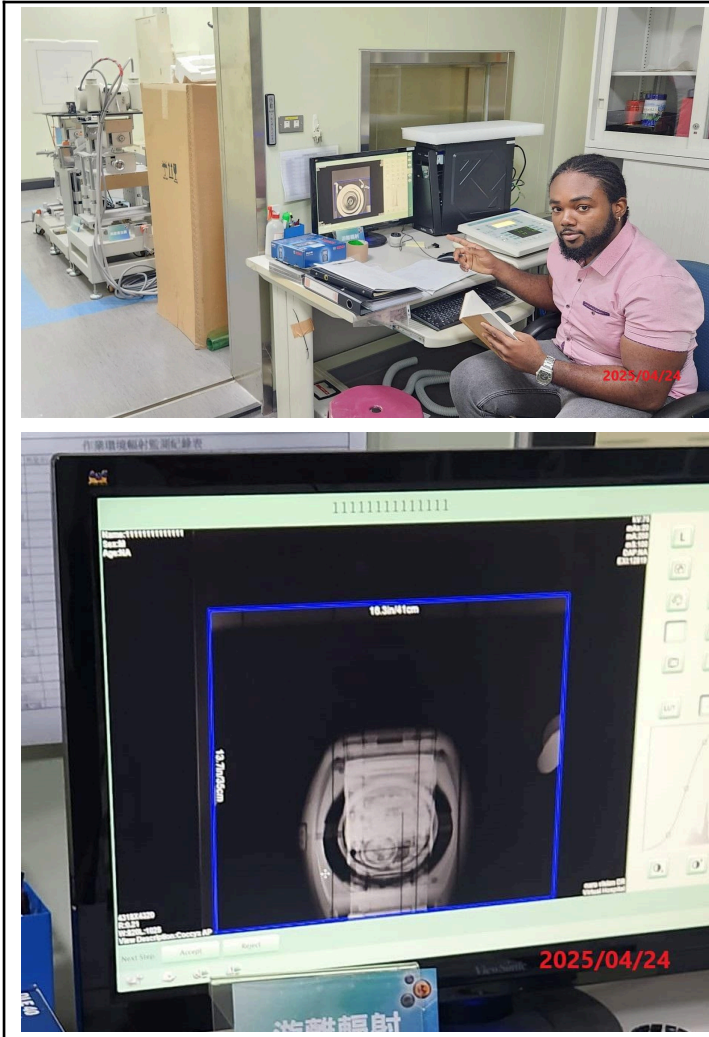
Here, we had the opportunity to test what we learned on a real CNC machine. The controls and movements were identical to the simulation; however, precision is critical when working with the actual machine, as mistakes can be costly. Safety remains the top priority when operating this equipment.



In this module, we studied two fundamental thin-film deposition techniques: PVD (Physical Vapor Deposition) and CVD (Chemical Vapor Deposition). PVD involves depositing a thin film onto a substrate through physical means. Typically by vaporizing the source material (via methods like sputtering or thermal evaporation) and allowing it to condense on the target surface without chemical reactions. In contrast, CVD relies on chemical reactions between gaseous precursors that decompose or react on the heated substrate surface to form the desired coating. While PVD offers better process control and higher purity, CVD generally provides superior step coverage and the ability to deposit more complex materials—though its reaction pathways can lead to lower reproducibility due to byproduct formation and sensitivity to process parameters.



This slide effectively outlines the Physical Vapor Transport (PVT) method for crystal growth, demonstrating a clear understanding of the temperature-driven sublimation and recrystallization process. The schematic of the seed crystal growth furnace, with its defined temperature zones ($\sim 2400^{\circ}\text{C}$ at the source and $\sim 2100^{\circ}\text{C}$ at the seed), visually reinforces the critical role of thermal gradients in transporting and depositing high-melting-point materials (e.g., Si, Si_2C , C_3).



X-ray machines detect defects in metallic products by passing radiation through the material and analyzing absorption patterns. Internal flaws like cracks, voids, or inclusions appear as lighter areas on radiographs due to reduced absorption compared to solid metal. This non-destructive method provides fast, accurate inspection of structural integrity without damaging components, making it essential for quality control in aerospace, automotive, and manufacturing industries. I got to test my Fossil watch as an example with this impressive analysis. Luckily, there were no signs of interior flaw.

Notes:

1. Please add rows to the table as needed. Each photo shall be dated at the bottom right, and each photo shall be accompanied by descriptions below it to explain the topic.
2. Please provide the original files of 1 to 3 photos (JPEG files, each above 1MB in size and with a resolution above 300 DPI).